Proofs of Theorems

Jeremy Rosen

1 Portfolio Choice with Matriculation

Theorem 1. *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. For brevity, let $w = (x, z)$. Then for large N, the portfolio choice problem is:

$$\max_{\delta \in \{\delta | \delta : \mathbb{R}^K \to [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w) f(w) dw \leq N_M, \tag{1}$$

where δ is the committee's decision rule, and f is the density of w. Because success and matriculation probabilities, as well as densities, are never negative, the unconstrained objective increases in δ, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}
\mathcal{L}(\delta; \lambda) &= \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw \\
&\quad + \lambda \left\{ N_M - \left[\int_{w \in \mathbb{R}^K} P(m = M|w) f(w) dw - \int_{w \in \mathbb{R}^K} P(m = M|w)(1 - \delta(w)) f(w) dw \right] \right\} \qquad (2) \\
&= \int_{w \in \mathbb{R}^K} \left[P(y = S|w)\delta(w) + \lambda(1 - \delta(w)) \right] P(m = M|w) f(w) dw + \lambda \left[N_M - \int_{w \in \mathbb{R}^K} P(m = M|w) f(w) dw \right].
\end{aligned}$$

In the discrete case, there is not necessarily a cutoff rule solution. But in the continuous case, the committee can accept the necessary fraction of applicants such that $\int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w|\lambda^*) f(w) dw = N_M$, which results in a cutoff rule solution on success probabilities alone:

$$\forall w \in \mathbb{R}^K,\ \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0,1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^*. \end{cases} \tag{3}$$

The committee accepts all applicants with success probabilities $P(y = S|w)$ strictly greater than the cutoff probability λ^*, rejects all applicants with $P(y = S|w)$ strictly less than λ^*, and may accept or reject any applicants with $P(y=S|w)$ equal to λ^*, if such a mass of applicants exists. All that remains is to identify λ^*. Like Bai et al. (2022), I define the following two functions:

$$\begin{aligned} M^H(\lambda) &= \int_{\{w\in\mathbb{R}^K|P(y=S|w)\geq\lambda\}} P(m=M|w)f(w)dw \\ M^L(\lambda) &= \int_{\{w\in\mathbb{R}^K|P(y=S|w)>\lambda\}} P(m=M|w)f(w)dw. \end{aligned} \tag{4}$$

The upper bound function M^H is the fraction of applicants, scaled by their matriculation probabilities, whose success probabilities are greater than or equal to some λ, and the lower bound function M^L is the fraction whose success probabilities are strictly greater than λ. Both functions are between 0 and 1, both are decreasing in λ, and for any λ, it must be that $M^H(\lambda) \geq M^L(\lambda)$.

Then it is the case that $\lambda^* = \inf_{\lambda}\{\lambda|M^L(\lambda) \leq N_M\}$. Intuitively, λ^* is the largest cutoff probability where the fraction of matriculation-probability-scaled applicants with $P(y=S|w) > \lambda$ does not exceed the maximum number of matriculants N_M. If there is a positive mass of matriculation-probability-scaled applicants with $P(y=S|w)=\lambda^*$ (that is, if $M^H(\lambda^*)-M^L(\lambda^*)>0$), then the committee will randomly accept the exact fraction $\alpha^* \in [0,1]$ of that mass of applicants such that $M^L(\lambda^*)+\alpha^*[M^H(\lambda^*)-M^L(\lambda^*)] = N_M$. That is, the committee will accept every applicant with $P(y=S|w) > \lambda^*$ and accept just enough applicants with $P(y=S|w)=\lambda^*$ to fill the cohort.

2 Cutoff Probability Identification

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^*=S|t)$ is such that the number of acceptances N_A equals the expected N_A. Also, for any $\hat{P}_{y^*}$ and N_A, $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n=S|x_n,z_n,t_n,\bar{k}_n)$.*

Proof. Supposing one round, the restricted likelihood is:

$$\mathcal{L}^R = \prod_{n=1}^{N}\left[\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma}+e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=A)}\left[\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma}+e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=R)}\left[\frac{e^{P_{y_n}}}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=S)}\left[\frac{1}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=F)} \tag{5}$$

When an applicant is accepted, the resulting term in $\mathcal{L}^R$ decreases in P_{y^*}, providing downward pressure on $\hat{P}_{y^*}$. Also, when one is rejected, the term increases in P_{y^*}, providing upward pressure.

Since $\mathbb{1}(d_n=A)+\mathbb{1}(d_n=R) = \mathbb{1}(y_n=S)+\mathbb{1}(y_n=F)=1$, the restricted log-likelihood is:

$$\begin{aligned} \ell^R = \sum_{n=1}^{N}&\left[\mathbb{1}(d_n=A)\log\left(\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma}+e^{P_{y^*}/\sigma}}\right)+\mathbb{1}(d_n=R)\log\left(\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma}+e^{P_{y^*}/\sigma}}\right)\right. \\ &\left.+\mathbb{1}(y_n=S)\log\left(\frac{e^{P_{y_n}}}{1+e^{P_{y_n}}}\right)+\mathbb{1}(y_n=F)\log\left(\frac{1}{1+e^{P_{y_n}}}\right)\right] \\ =\sum_{n=1}^{N}&\left[\frac{1}{\sigma}\left(\mathbb{1}(d_n=A)P_{y_n}+\mathbb{1}(d_n=R)P_{y^*}\right)-\log\left(e^{P_{y_n}/\sigma}+e^{P_{y^*}/\sigma}\right)+\mathbb{1}(y_n=S)P_{y_n}-\log\left(1+e^{P_{y_n}}\right)\right]. \end{aligned} \tag{6}$$

Since ℓ^R is strictly concave in P_{y^*}, I find the unique maximizer $\hat{P}_{y^*}$ by the first-order condition:

$$\frac{1}{\sigma}\left[N - N_A - \sum_{n=1}^{N} \frac{e^{\hat{P}_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}}\right] = 0 \quad \Rightarrow \quad \sum_{n=1}^{N} \frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} = N_A. \tag{7}$$

The left side of the latter equation is each applicant's acceptance probability summed over the number of applicants; that is, expected N_A. Therefore, $\hat{P}_{y^*}$ is such that the expected N_A equals N_A.

Given that $\hat{P}_{y^*}$ is identified, I will show that ℓ^R, and therefore $\mathcal{L}^R$, are highest when the committee accepts the N_A applicants with the highest P_{y_n}. The only part of ℓ^R that depends on d_n is $\sum_{n=1}^{N}[\mathbb{1}(d_n = A)P_{y_n} + \mathbb{1}(d_n = R)P_{y^*}] = NP_{y^*} + \sum_{n=1}^{N}\mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$; this equality holds since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = 1$. As such, the only part depending on d_n is $\sum_{n=1}^{N}\mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$.

This expression is highest when $d_n = A$ for the N_A highest values of P_{y_n}, so the likelihood is highest when the committee accepts the N_A highest P_{y_n}. Intuitively, the committee's selectivity identifies $\hat{P}_{y^*}$, and it is not necessary for identification for the committee to accept applicants with higher P_{y_n}. However, if the committee engages in positive sorting, the model's fit will improve.

3 Randomization Test P-Value

Theorem 3. *For large B, the fraction of $\tilde{\theta}^U_{S,b}$ with gain $g(\tilde{\theta}^U_{S,b}) \leq 0$ is a p-value for $H_0\colon g_0 \leq 0$.*

Proof. Define deviation gain $g(\theta_S) = \frac{1}{N_A}\sum_{n=1}^{N} P\Big(y_n = S\Big| x_n, z_n, t_n, \bar{k}_n; \theta_S\Big)\mathbb{1}(d^R_{1,n} = T, d^R_{2,n} = A) - \frac{1}{N_A}\sum_{n=1}^{N} P\Big(y_n = S\Big| x_n, z_n, t_n, \bar{k}_n; \theta_S\Big)\mathbb{1}(d^U_{1,n} = T, d^U_{2,n} = A)$. That is, the deviation gain is the average success probability of the model's acceptance set minus that of the committee's acceptance set. Since the model and committee's sets are determined before performing the test, the proof works the same way for matriculation sets as it does for acceptance sets. Suppose that the maximum likelihood estimate $\hat{\theta}^U_S \overset{a}{\sim} \mathcal{N}(\theta^0_S, \Sigma)$, and that each perturbation draw $\tilde{\theta}^U_{S,b} \sim \mathcal{N}(\hat{\theta}^U_S, \Sigma)$.

Because g is a smooth function of θ_S, it is true by the delta method that $\hat{g} = g(\hat{\theta}^U_S) \overset{a}{\sim} \mathcal{N}(g_0, \sigma^2)$ and $\tilde{g} = g(\tilde{\theta}^U_{S,b}) \mid \hat{g} \overset{a}{\sim} \mathcal{N}(\hat{g}, \sigma^2)$, where $g_0 = g(\theta^0_S)$ and $\sigma^2 = \nabla g(\theta^0_S)'\Sigma\nabla g(\theta^0_S)$. As the number of perturbations B becomes large, it is true by the law of large numbers that the fraction of perturbations with $\tilde{g} \leq 0$ converges to the probability that a single perturbation is weakly negative:

$$\hat{p} = P(\tilde{g} \leq 0 \mid \hat{g}) = \Phi\left(\frac{0 - \hat{g}}{\sigma}\right) = \Phi(-\hat{g}/\sigma). \tag{8}$$

For $\hat{p}$ to be a p-value, it must equal the probability, if the null is true with $g_0 = 0$, of an estimated gain at least as large as the observed gain. Denote $\hat{g}' = \sigma Z$ as a hypothetical estimated gain generated under $g_0 = 0$, where $Z \sim \mathcal{N}(0,1)$. Then, using $P(Z \geq c) = \Phi(-c)$:

$$P(\hat{g}' \geq \hat{g} \mid g_0 = 0) = P(Z \geq \hat{g}/\sigma) = \Phi(-\hat{g}/\sigma) = \hat{p}. \tag{9}$$

That is, the probability, if the null is true with $g_0 = 0$, of an estimated gain at least as large as the observed gain equals the probability to which the fraction of perturbations with $\tilde{g} \leq 0$ converges.

Finally, I show that whenever $g_0 \leq 0$, the probability of a Type 1 error is bounded above by the statistical significance threshold α. That is, $P(\hat{p} \leq \alpha) \leq \alpha$ whenever $g_0 \leq 0$. Because inverses of strictly increasing functions preserve inequalities, Φ is strictly increasing, and $\hat{p} = \Phi(-\hat{g}/\sigma)$; it is the case that $\hat{p} \leq \alpha$ holds exactly when $\hat{g} \geq -\sigma\Phi^{-1}(\alpha)$. Substituting $\hat{g} = g_0 + \sigma Z$:

$$P(\hat{p} \leq \alpha) = P(Z \geq -\Phi^{-1}(\alpha) - g_0/\sigma) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma). \tag{10}$$

For $g_0 = 0$, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha)) = \alpha$. For $g_0 < 0$, because Φ is strictly increasing, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma) < \alpha$. Therefore, rejecting H_0 when $\hat{p} \leq \alpha$ produces Type 1 errors with probability at most α everywhere that H_0 is true.

4 Portfolio Choice with Shocks

Theorem 4. *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. The portfolio choice problem with type 1 extreme value preference shocks is:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \left[(P(y_n = S|x_n,z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + \varepsilon_n(R)\mathbb{1}(d_n = R)\right], \text{ such that } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \tag{11}$$

Because success probabilities are never negative, the unconstrained objective increases in expectation every time $d_n = A$, and so the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N} \left[(P(y_n = S|x_n,z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + \varepsilon_n(R)\mathbb{1}(d_n = R)\right] + \lambda\left\{N_A - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_n = R)\right]\right\} \\ &= \sum_{n=1}^{N} \left[(P(y_n = S|x_n,z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + (\lambda + \varepsilon_n(R))\mathbb{1}(d_n = R)\right] + \lambda(N_A - N). \end{aligned} \tag{12}$$

Suppose without loss of generality that the applicants are sorted in decreasing order of their shock-perturbed success probabilities such that $P(y_1 = S|x_1,z_1) + \varepsilon_1(A) - \varepsilon_1(R) \geq \cdots \geq P(y_N = S|x_N,z_N) + \varepsilon_N(A) - \varepsilon_N(R)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^* = \lambda^*(\boldsymbol{\varepsilon}(A),\boldsymbol{\varepsilon}(R)) \in \left[P(y_{N_A+1} = S|x_{N_A+1},z_{N_A+1}) + \varepsilon_{N_A+1}(A) - \varepsilon_{N_A+1}(R), P(y_{N_A} = S|x_{N_A},z_{N_A}) + \varepsilon_{N_A}(A) - \varepsilon_{N_A}(R)\right]$. That is, in this setup, λ^* is set identified. Then the solution is to accept all applicants whose shock-perturbed success probability exceeds λ^* and reject the rest:

$$\forall n \in \{1,\ldots,N\},\, d_n(x_n,z_n,\varepsilon_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n) + \varepsilon_n(A) > \lambda^* + \varepsilon_n(R) \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n) + \varepsilon_n(A) = \lambda^* + \varepsilon_n(R) \\ R & \text{if } P(y_n = S|x_n,z_n) + \varepsilon_n(A) < \lambda^* + \varepsilon_n(R). \end{cases} \tag{13}$$

This cutoff rule holds conditional on the realized shocks. Given the shocks, the committee computes λ^* and applies the cutoff, so there is no difficulty for the committee. But there is a complication regarding estimation, namely that unlike the committee, the econometrician does not observe the shocks. Therefore, the conditional choice probabilities used in estimation require taking expectations across each applicant's shocks, which would be fine except λ^* is a function of each applicant's shocks simultaneously. Suppose for now that λ^* is instead deterministic. Then:

$$\begin{aligned} P(d_n = A|x_n, z_n) &= P(P(y_n = S|x_n, z_n) + \varepsilon_n(A) \geq \lambda^* + \varepsilon_n(R)) \\ &= P(\varepsilon_n(R) - \varepsilon_n(A) \leq P(y_n = S|x_n, z_n) - \lambda^*) \\ &= \frac{\exp(P(y_n = S|x_n, z_n)/\sigma)}{\exp(P(y_n = S|x_n, z_n)/\sigma) + \exp(\lambda^*/\sigma)}, \end{aligned} \tag{14}$$

and the likelihood becomes the product of individual binary logistics. The third equality holds because the difference of two independent T1EV random variables has a logistic distribution.

That said, λ^* is not deterministic for finite N, so the logistic conditional choice probabilities do not hold exactly. But for large N, λ^* converges to a constant. Denote $\hat{F}_N(c) = \frac{1}{N}\sum_{n=1}^{N} \mathbb{1}\{P(y_n = S|x_n, z_n) + \varepsilon_n(A) - \varepsilon_n(R) \leq c\}$ as the empirical distribution of the shock-perturbed success probabilities. The cutoff λ^* is the $(1 - N_A/N)$ quantile of this distribution. Because the shocks are iid, the perturbed indices in $\hat{F}_N$ are iid. So by the Glivenko-Cantelli theorem, $\hat{F}_N$ converges almost surely to the population distribution F for large N. Therefore, as $N_A/N \to q$, $\hat{F}_N$'s $(1 - N_A/N)$ quantile converges to F's deterministic $(1 - q)$ quantile. The iid assumption is essential, and not only because each applicant's shock individually moves the cutoff by $O(1/N)$. A common perturbation component across applicants would have the same vanishing per-applicant influence, but it would slide the entire empirical distribution in unison, leaving λ^* random in the limit. Independence makes the simultaneous fluctuations of the N shocks cancel, so $\hat{F}_N$ and λ^* converge. With λ^* deterministic, no individual applicant's shock affects λ^*, the expectation defining each conditional choice probability factors across applicants, and the above logistic form holds.

5 Deviation Gains

Theorem 5. *Let d^R be the model's decision rule with $\sigma = 0$ in the CCPs, and let d^U be the committee's decision rule. By optimality:*

$$\begin{aligned} &\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\} \\ &\qquad \geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\} \end{aligned} \tag{15}$$

Proof. This theorem states that conditional on a fixed Round 1 decision rule $d_1[P_y(\hat{\theta}^U)]$ and corresponding transition set of exogenously-determined size N_T, the sample average success probability of the also-exogenous N_A applicants that the model accepts in Round 2 weakly exceeds the sample average success probability of the N_A applicants that any actual committee accepts in Round 2. The inequality holds since the model's Round 2 decision rule is to accept the N_A applicants with the highest success probabilities. Therefore, the sample average success probability of the model's acceptances must weakly exceed that of any other set of N_A acceptances.

However, the inequality would not always hold if it were written as follows (changes in bold):

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S\middle|x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right)\mathbb{1}\left\{\mathbf{d^R_{1,n}}\left[\mathbf{P_{y_n}}\left(\hat{\theta}^{\mathbf{U}}_{\mathbf{S}}\right)\right] = \mathbf{T}, d^R_{2,n}\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$\textbf{vs. } \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S\middle|x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right)\mathbb{1}\left\{\mathbf{d^U_{1,n}}\left[\mathbf{P_{y_n}}\left(\hat{\theta}^{\mathbf{U}}_{\mathbf{T}}\right)\right] = \mathbf{T}, d^U_{2,n}\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\} \quad (16)$$

The difference is that the Round 1 transition set is no longer fixed. Now, the model's transition set is based on choosing the N_T applicants in each year with the highest $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma})dF(\tilde{z}|x,t)$. In contrast, the committee may have a different Round 1 decision rule. Crucially, although the model is choosing applicants akin to those with the highest expected success probabilities, it is ignoring any value added from upside potential.

For example, suppose $N_A = 1$, and there is an applicant in Round 1 with a low expected success probability but a known high degree of variance in the distribution of z conditional on their (x_n, t_n). That is, $F(z|x,t)$ is heteroskedastic. Also suppose that their low expected success probability yields an $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$ too low to survive Round 1, even though their actual success probability ends up being the highest of any applicant once z is realized in Round 2. If the committee's decision rule considers upside potential, then the committee may accept this applicant, while the model will not. In most cases, the model's Round 1 decision rule should lead to a better acceptance set than the committee's rule. But such an outcome is not guaranteed.

6 Waitlist Models

Theorem 6. *In the waitlist-dynamic and hybrid models, success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y=S|x^H,z^H,t,\bar{k}) = P(y=S|x^L,z^L,t,\bar{k})$, but $m_2^H > m_2^L$. That is, the applicants have equal success probabilities, but H has the higher matriculation probability. Since $\ddot{m}_2$ is the average matriculation probability of all applicants in t with m_2 excluded, $\ddot{m}_2^H < \ddot{m}_2^L$ by construction. Given this inequality, I will show $P(d_2 = A|x^H,z^H,t,\bar{k},\ddot{m}_2^H) > P(d_2 = A|x^L,z^L,t,\bar{k},\ddot{m}_2^L)$. It is the case that:

$$P(d_2 = A|x,z,t,\bar{k},\ddot{m}_2) = \frac{\exp(P(y=S|x,z,t,\bar{k})/\sigma)}{\exp(P(y=S|x,z,t,\bar{k})/\sigma) + \exp(\beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2]/\sigma)}, \text{ where}$$
$$\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2). \quad (17)$$

The only place $\ddot{m}_2$ appears in $P(d_2 = A|x,z,t,\bar{k},\ddot{m}_2)$ is in the distribution of $\overline{m}_{2A}$ conditional on $\ddot{m}_2$. I parameterize $\overline{m}_{2A}|\ddot{m}_2 \sim \mathcal{N}(\ddot{m}_2\beta_{\ddot{m}_2}, \sigma^2_{\ddot{m}_2})$, where by assumption $\beta_{\ddot{m}_2} > 0$. $\beta_{\ddot{m}_2}$ is positive because years when $\ddot{m}_2$ is high are years when matriculation probabilities are high in general.

Using inter-year variation, this relationship results in a greater expected $\overline{m}_{2A}$ in those years. Then:

$$F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^H;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2}) = \Phi\left(\frac{\overline{m}_{2A}-\ddot{m}_2^H\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right) > \Phi\left(\frac{\overline{m}_{2A}-\ddot{m}_2^L\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right) = F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^L;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2}). \tag{18}$$

Therefore, $\overline{m}_{2A}|\ddot{m}_2^L$ first-order stochastically dominates $\overline{m}_{2A}|\ddot{m}_2^H$. To apply this stochastic dominance, I must show that $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$ is strictly increasing in $\overline{m}_{2A}$. But the only place $\overline{m}_{2A}$ appears in this expression is in $P(y_3^*=S|\overline{m}_{2A};\theta_{S_3^*}) = \frac{\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}{1+\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}$, where by assumption $f_{S_3^*}(\cdot)$ is strictly increasing and $\theta_{S_3^*}>0$. $\theta_{S_3^*}$ is positive because the expected Round 3 cutoff probability is higher when early acceptees are more likely on average to matriculate. As such, I have that $P(y_3^*=S|\overline{m}_{2A})$, and in turn $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$, are strictly increasing in $\overline{m}_{2A}$. Combining this result with $(\overline{m}_{2A}|\ddot{m}_2^L)\succ_{\text{FOSD}}(\overline{m}_{2A}|\ddot{m}_2^H)$, I obtain:

$$\begin{aligned}
\mathbb{E}[v_3(x^L,z^L,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x^L,z^L,t,\bar{k},\ddot{m}_2^L] &= \int_{\widetilde{\overline{m}_{2A}}} \sigma\log\left(e^{P(y=S|x^L,z^L,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L)\\
&= \int_{\widetilde{\overline{m}_{2A}}} \sigma\log\left(e^{P(y=S|x^H,z^H,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L)\\
&> \int_{\widetilde{\overline{m}_{2A}}} \sigma\log\left(e^{P(y=S|x^H,z^H,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^H)\\
&= \mathbb{E}[v_3(x^H,z^H,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x^H,z^H,t,\bar{k},\ddot{m}_2^H].
\end{aligned} \tag{19}$$

The second equality holds since $P(y=S|x^H,z^H,t,\bar{k})=P(y=S|x^L,z^L,t,\bar{k})$. Therefore, by Equation (17) and Inequality (19), I have that $P(d_2=A|x^H,z^H,t,\bar{k},\ddot{m}_2^H)>P(d_2=A|x^L,z^L,t,\bar{k},\ddot{m}_2^L)$.

7 Multiyear Model (Matriculation)

Theorem 7. *In the matriculation multiyear model, success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y=S|x^H,z^H,t)=P(y=S|x^L,z^L,t)$, but $m^H>m^L$. That is, H has the higher matriculation probability. I will show that $P(d_2=A|x^H,z^H,t,m^H,c_{-1})>P(d_2=A|x^L,z^L,t,m^L,c_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\begin{aligned}
\pi_2^A &\equiv P(y=S|x,z,t)+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m], \quad \pi_2^R \equiv L(y_2^*=S|c_{-1})+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1')],\\
\pi_1^A &\equiv \mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c], \quad \pi_1^R \equiv L(y_1^{*\prime}=S|t')+\mathbb{E}[v_1(x'',t'',c',\varepsilon_1'')].
\end{aligned} \tag{20}$$

It is the case that:

$$P(d_2=A|x,z,t,m,c_{-1}) = \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma)+\exp(\pi_2^R/\sigma)}, \text{ where:}$$

$$\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m] = \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m), \text{ where:}$$

$$\mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c] = \int_{\widetilde{m'}} \int_{\widetilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\widetilde{z'}|x',t') dF_{m|x,t}(\widetilde{m'}|x',t'), \quad (21)$$

where $\pi_2^{A\prime}$ and $\pi_2^{R\prime}$ are defined analogously to Equation (20) at the next year's states.

Per-period utilities are bounded and $\beta \in (0,1)$, so the value functions exist and are unique. The only place m appears in $P(d_2 = A|x,z,t,m,c_{-1})$ is in the distribution of c conditional on m. I parameterize $c|m \sim \mathcal{N}(m\beta_m, \sigma_m^2)$, where by assumption $\beta_m > 0$. β_m is positive because higher matriculation probabilities lead to a greater expected cohort size. Multiple years of data are necessary to identify β_m, since cohort sizes vary only across years. Then:

$$F_{c|m}(c|m^H;\beta_m,\sigma_m^2) = \Phi\left(\frac{c-m^H\beta_m}{\sigma_m}\right) < \Phi\left(\frac{c-m^L\beta_m}{\sigma_m}\right) = F_{c|m}(c|m^L;\beta_m,\sigma_m^2). \quad (22)$$

Therefore, $c|m^H$ first-order stochastically dominates $c|m^L$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m]$'s integrand consisting of all but the outermost integral is strictly increasing in c. But the only place c appears there is in $\mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c]$, and the only place c appears in that expectation is in $L(y_2^{*\prime} = S|c;\theta_{S_2^*}) = f_{S_2^*}(c)\theta_{S_2^*}$.

By assumption, $f_{S_2^*}(\cdot)$ is strictly increasing and $\theta_{S_2^*} > 0$. $\theta_{S_2^*,c}$ is positive because a greater cohort size this year means that the committee, needing to keep the total number of students in the program relatively constant across years, can afford to be more selective next year. As such, I have that $P(y_2^{*\prime} = S|c)$, and in turn $\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m]$'s integrand, are strictly increasing in c. Combining this result with $[c|m^H] \succ_{FOSD} [c|m^L]$, I obtain the inequality:

$$\begin{aligned}
\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m^H] &= \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m^H) \\
&> \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m^L) \\
&= \mathbb{E}[v_1(x',t',c,\varepsilon_1')|m^L].
\end{aligned} \quad (23)$$

Therefore, by Equation (21), Inequality (23), and $P(y = S|x^H,z^H,t) = P(y = S|x^L,z^L,t)$, I have that $P(d_2 = A|x^H,z^H,t,m^H,c_{-1}) > P(d_2 = A|x^L,z^L,t,m^L,c_{-1})$. Intuitively, if the committee rejects either applicant, their m does not matter. But since their success probabilities are equal, accepting applicant H yields a higher expected payoff than accepting applicant L.

8 Multiyear Model (Diversity)

Theorem 8. *Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z. Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p, the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.*

Proof. Suppose there are two arbitrary applicants in year t, one with $a = 1$ and the other with $a = 0$, but with $P(y = S|x^1, z^1, t, 1-\bar{a}_{-1}) = P(y = S|x^0, z^0, t, \bar{a}_{-1})$. Unlike in the previous two proofs, these equal success probabilities imply that unless $\bar{a}_{-1} = \frac{1}{2}$, it must be that $(x^1, z^1) \neq (x^0, z^0)$. I will show that $P(d_2 = A|x^1, z^1, t, 1, \bar{a}_{-1}) > P(d_2 = A|x^0, z^0, t, 0, \bar{a}_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\pi_2^A \equiv P(y = S|x, z, t, |a-\bar{a}_{-1}|) + \beta\mathbb{E}[v_1(x', t', \bar{a}, \varepsilon_1')|a], \quad \pi_2^R \equiv L(y_2^* = S|t) + \beta\mathbb{E}[v_1(x', t', \bar{a}, \varepsilon_1')],$$

$$\pi_1^A \equiv \mathbb{E}[v_2(x', z', t', a', \bar{a}, \varepsilon_2')|x', t', \bar{a}], \quad \pi_1^R \equiv L(y_1^{*\prime} = S|t') + \beta\mathbb{E}[v_1(x'', t'', \bar{a}', \varepsilon_1'')]. \tag{24}$$

It is the case that:

$$P(d_2 = A|x, z, t, a, \bar{a}_{-1}) = \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma) + \exp(\pi_2^R/\sigma)}, \text{ where:}$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \varepsilon_1')|a] = \int_{\tilde{\bar{a}}}\int_{\tilde{t'}}\int_{\tilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\tilde{x'})\, dF_t(\tilde{t'})\, dF_{\bar{a}|a}(\tilde{\bar{a}}|a), \text{ where:}$$

$$\begin{aligned}\mathbb{E}[v_2(x', z', t', a', \bar{a}, \varepsilon_2')|x', t', \bar{a}] = P(a' = 1)\int_{\tilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}(1)/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\tilde{z'}|x', t') \\ + [1 - P(a' = 1)]\int_{\tilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}(0)/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\tilde{z'}|x', t'),\end{aligned} \tag{25}$$

where $\pi_2^{A\prime}(a')$ denotes next year's Round 2 acceptance payoff evaluated at $a' \in \{0, 1\}$.

Per-period utilities are bounded and $\beta \in (0, 1)$, so the value functions exist and are unique. The only two places a appears in $P(d_2 = A|x, z, t, a, \bar{a}_{-1})$ are in the success probabilities, which are assumed to be equal for the $a = 1$ and $a = 0$ applicants, and in the distribution of $\bar{a}$ conditional on a. I parameterize $\bar{a}|a \sim \mathcal{N}(a\beta_a, \sigma_a^2)$, where by assumption $\beta_a > 0$. β_a is positive because $a = 1$ applicants are more likely than $a = 0$ applicants to be in a year with a high percentage of accepted $a = 1$ applicants. As such, multiple years of data are necessary to identify β_a. Then:

$$F_{\bar{a}|a}(\bar{a}|1; \beta_a, \sigma_a^2) = \Phi\left(\frac{\bar{a} - \beta_a}{\sigma_a}\right) < \Phi\left(\frac{\bar{a}}{\sigma_a}\right) = F_{\bar{a}|a}(\bar{a}|0; \beta_a, \sigma_a^2). \tag{26}$$

Therefore, $\bar{a}|1$ first-order stochastically dominates $\bar{a}|0$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x', t', \bar{a}, \varepsilon_1')|a]$'s integrand consisting of all but the outermost integral is strictly increasing in $\bar{a}$. But the only place $\bar{a}$ appears there is in $\mathbb{E}[v_2(x', z', t', a', \bar{a}, \varepsilon_2')|x', t', \bar{a}]$, and the only place $\bar{a}$ appears in that expectation is in $P(y' = S|x', z', t', \bar{a}; \theta_S) = \frac{\exp(f_S(x', z', t', |a'-\bar{a}|)\theta_S)}{1+\exp(f_S(x', z', t', |a'-\bar{a}|)\theta_S)}$.

By assumption, $f_S(\cdot)$ is strictly increasing and $\theta_{S,|a-\bar{a}_{-1}|} > 0$. $\theta_{S,|a-\bar{a}_{-1}|}$ is positive because a large value of $|a - \bar{a}_{-1}|$ means that applicant a is diverse relative to the previous year's accepted applicants and is therefore more likely to succeed. Recall $P(a' = 1) = p$. Taking the limit $p \to 0$:

$$\begin{aligned}&\mathbb{E}[v_2(x', z', t', a', \bar{a}, \varepsilon_2')|x', t', \bar{a}] \\ &\to \int_{\tilde{z'}} \sigma \log(e^{\{P(y'=S|x', z', t', \bar{a}) + \beta\mathbb{E}[v_1(x'', t'', \bar{a}', \varepsilon_1'')|0]\}/\sigma} + e^{\{L(y_2^{*\prime}=S|t') + \beta\mathbb{E}[v_1(x'', t'', \bar{a}', \varepsilon_1'')]\}/\sigma}) dF_{z|x,t}(\tilde{z'}|x', t').\end{aligned} \tag{27}$$

$P(y' = S|x', z', t', \overline{a})$ is strictly increasing in $\overline{a}$, and by the continuity of $E[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}]$ in p, I have that it, and in turn $E[v_1(x', t', \overline{a}, \varepsilon_1')|a]$'s integrand, are also strictly increasing in $\overline{a}$ for sufficiently small p. Combining this result with $\overline{a}|1 \succ_{FOSD} \overline{a}|0$, I obtain the inequality:

$$\begin{aligned}\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|1] &= \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|1) \\ &> \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|0) \\ &= \mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|0].\end{aligned} \tag{28}$$

Therefore, by Equation (25), Inequality (28), and $P(y = S|x^1, z^1, t, 1-\overline{a}_{-1}) = P(y = S|x^0, z^0, t, \overline{a}_{-1})$, I have that $P(d_2 = A|x^1, z^1, t, 1, \overline{a}_{-1}) > P(d_2 = A|x^0, z^0, t, 0, \overline{a}_{-1})$. Intuitively, if the committee rejects either applicant, their a does not matter. But since their success probabilities are equal, accepting applicant $a = 1$ yields a higher expected payoff than accepting $a = 0$.

In this proof, I must assume $P(a' = 1)$ is sufficiently small, and not just $P(a' = 1) < \frac{1}{2}$. The reason is that if $P(a' = 1)$ is too large, then $\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}]$ may not be strictly increasing in $\overline{a}$, considering $P(y' = S|x', z', t', 1-\overline{a})$ is strictly decreasing in $\overline{a}$. This indeterminacy can be explored by calculating $\frac{\partial}{\partial \overline{a}}\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}]$. As such, to be sure the committee is more likely to accept the $a = 1$ applicant, the committee must know that there will be sufficiently few $a' = 1$ applicants. This assumption is strong, but possible if $a' = 1$ is sufficiently rare for the committee to know this will happen. However, even without forward-looking dynamics, if $\overline{a}_{-1} < \frac{1}{2}$, then the committee will be more likely to accept an $a = 1$ applicant than an $a = 0$ applicant with identical x and z values. The reason is that $P(y = S|x, z, t, 1-\overline{a}_{-1}) > P(y = S|x, z, t, \overline{a}_{-1})$.